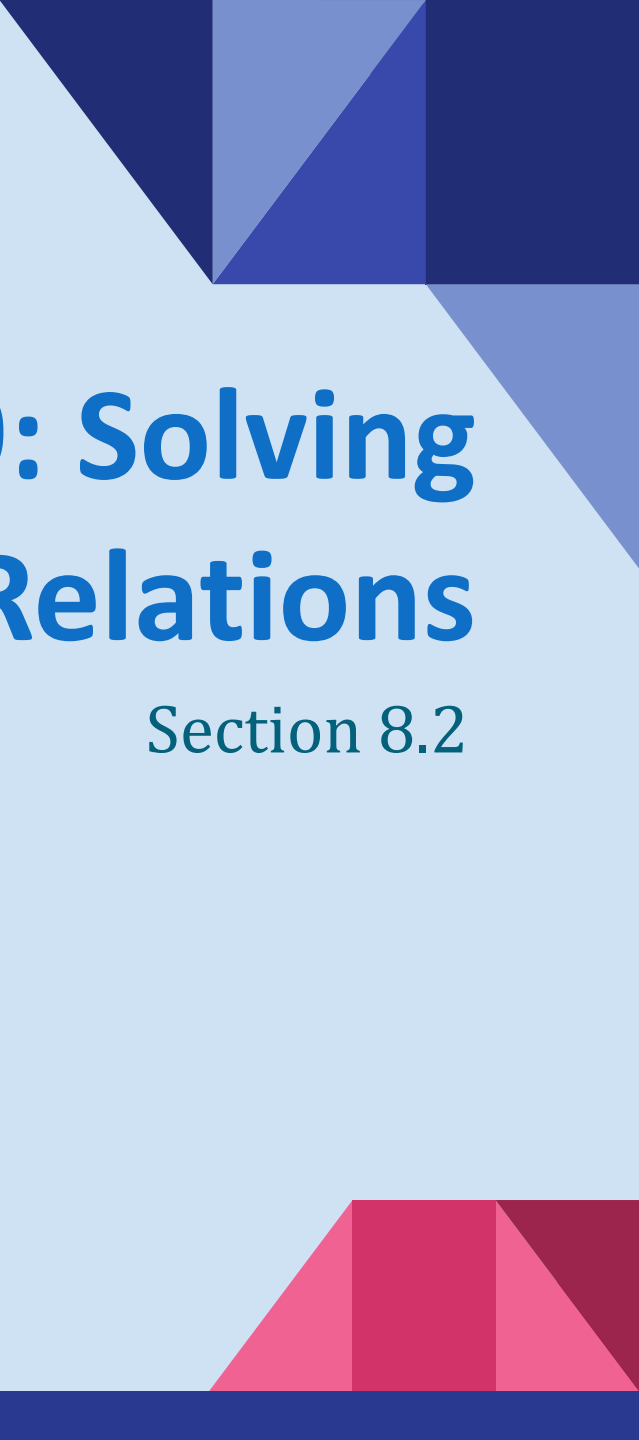


The slide features a light blue background. In the top right corner, there is a decorative arrangement of overlapping triangles in various shades of blue and purple. In the bottom right corner, there is a similar arrangement of overlapping triangles in shades of pink and red. A solid dark blue horizontal bar runs across the bottom of the slide.

Lecture 9: Solving Linear Recurrence Relations

Section 8.2

Section Summary

- Linear Homogeneous Recurrence Relations
- Solving Linear Homogeneous Recurrence Relations with Constant Coefficients.
- Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients. (Self Study)

Linear Homogeneous Recurrence Relations

Definition: A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$, where $c_1, c_2, \dots, c_k$ are real numbers, and $c_k \neq 0$.

- It is *linear* because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of n .
- It is *homogeneous* because no terms occur that are not multiples of the recursive elements.
- Each coefficient is a constant.
- The *degree* is k because a_n is expressed in terms of the previous k terms of the sequence.

Examples of Linear Homogeneous Recurrence Relations

• $P_n = (1.11)P_{n-1}$	linear homogeneous recurrence relation of degree one
• $f_n = f_{n-1} + f_{n-2}$	linear homogeneous recurrence relation of degree two
• $a_n = a_{n-1} + a_{n-2}^2$	not linear
• $H_n = 2H_{n-1} + 1$	not homogeneous
• $B_n = nB_{n-1}$	coefficients are not constants
• $a_n = a_{n-5}$	linear homogeneous recurrence relation of degree five

Solving Linear Homogeneous Recurrence Relations

- The basic approach is to look for solutions of the form

$$a_n = r^n, \text{ where } r \text{ is a constant.}$$

- Notice that $a_n = r^n$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

$$\text{if and only if } r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}.$$

- Divide both sides of this equation by r^{n-k} .

$$r^k = c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_{k-1} r + c_k$$

Solving Linear Homogeneous Recurrence Relations

- Algebraic manipulation yields the characteristic equation:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_{k-1} r - c_k = 0$$

- The sequence $\{a_n\}$ with $a_n = r^n$ is a solution if and only if r is a solution to this equation.
- The solutions to the characteristic equation are called the *characteristic roots* of the recurrence relation. The roots are used to give an explicit formula for all the solutions of the recurrence relation.

Solving Linear Homogeneous Recurrence Relations

- Another observation:

A **linear combination** of two solutions is also a solution.

So, if the recurrence $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ have s_n and t_n both as valid solutions, so,

$$s_n = c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k} \text{ and}$$
$$t_n = c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}.$$

Consider a linear combination of s_n and t_n : $b_1 s_n + b_2 t_n$, where b_1 and b_2 are real numbers.

Solving Linear Homogeneous Recurrence Relations

We have:

$$\begin{aligned} & b_1 s_n + b_2 t_n \\ &= b_1(c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}) + b_2(c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}) \\ &= c_1(b_1 s_{n-1} + b_2 t_{n-1}) + c_2(b_1 s_{n-2} + b_2 t_{n-2}) + \cdots + c_k(b_1 s_{n-k} + b_2 t_{n-k}). \end{aligned}$$

- So, if s_n and t_n both are solutions, then $b_1 s_n + b_2 t_n$ is also a solution.
- Combining that with the idea solutions can be of the form r^n , gives us the following theorem:

Solving Linear Homogeneous Recurrence Relations of Degree Two

Let c_1 and c_2 be real numbers.

Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2 .

Now, the degree-2 recurrence relation, $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ has the solution of the form: $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

THEOREM 1

Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Example: What is the solution to the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2} \text{ with } a_0 = 2 \text{ and } a_1 = 7?$$

Solution:

- **Step-1: find the *characteristic equation***

For our degree 2 recurrence relations, the characteristic equation is $r^2 - r - 2 = 0$. ($c_1 = 1, c_2 = 2$)

- **Step-2: find the *roots of the characteristic equation***

Its roots are $r = 2$ and $r = -1$.

- **Step-3: find the *general solution* format with α_1 and α_2**

Therefore, $\{a_n\}$ is a solution if and only if $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, for some constants α_1 and α_2 .

- **Step-4: find the values of α_1 and α_2 using **the initial conditions****

$$a_0 = 2 = \alpha_1 + \alpha_2 \text{ and } a_1 = 7 = \alpha_1 2 + \alpha_2 (-1).$$

Solving these equations, we find that $\alpha_1 = 3$ and $\alpha_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

An Explicit Formula for the Fibonacci Numbers

We can find an explicit formula for the Fibonacci numbers. The sequence of Fibonacci numbers satisfies the recurrence relation: $f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

Solution : The roots of the characteristic equation $r^2 - r - 1 = 0$ are:

$$r_1 = \frac{1+\sqrt{5}}{2}, \quad r_2 = \frac{1-\sqrt{5}}{2}.$$

So, for some constants α_1 and α_2 ,

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Using the initial conditions:

$$f_0 = \alpha_1 + \alpha_2 = 0; \quad f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1$$

We solve these equations to find: $\alpha_1 = \frac{1}{\sqrt{5}}, \quad \alpha_2 = -\frac{1}{\sqrt{5}}$

Thus,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

What if there is a repeated root from the characteristic equation?

Let c_1 and c_2 be real numbers with $c_2 \neq 0$.

Suppose that $r^2 - c_1 r - c_2 = 0$ has one repeated root r_0 .

Then the sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

THEOREM 2

Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0 . A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

Example Problem

Example : What is the solution to the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} \text{ with } a_0 = 1 \text{ and } a_1 = 6?$$

Solution : The characteristic equation is $r^2 - 6r + 9 = 0$.

The only root is $r = 3$.

Therefore, $\{a_n\}$ is a solution if and only if $a_n = \alpha_1 3^n + \alpha_2 n(3)^n$,

where α_1 and α_2 are constants.

From the given initial conditions,

$$a_0 = 1 = \alpha_1 \quad \text{and,} \quad a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 1 \cdot 3.$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Hence, $a_n = 3^n + n3^n$.

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

This theorem can be used to solve linear homogeneous recurrence relations with constant coefficients of any degree when the characteristic equation has distinct roots.

Theorem 3: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has k distinct roots $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n.$$

for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example Problem

Example : What is the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3} \text{ with } a_0 = 2, a_1 = 5, a_2 = 15?$$

Solution : The characteristic equation is $r^3 - 6r^2 + 11r - 6 = 0$.

The roots are, $r_1 = 1, r_2 = 2, r_3 = 3$.

Therefore, the solution is of the form $a_n = \alpha_1 1^n + \alpha_2 2^n + \alpha_3 3^n$, where α_1, α_2 and α_3 are constants.

From the given initial conditions,

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3, \quad a_1 = 5 = \alpha_1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3$$

and

$$a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = -1$ and $\alpha_3 = 2$.

Hence, $a_n = 1 - 2^n + 2 \cdot 3^n$.

The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, r_2, \dots, r_t$ with multiplicities $m_1, m_2, \dots, m_t$ respectively so that $m_i \geq 1$ for $i = 1, 2, \dots, t$ and $m_1 + m_2 + \dots + m_t = k$.

Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

if and only if $a_n = (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n$
 $+ (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n$
 $+ \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n$

for $n = 0, 1, 2, \dots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_{i-1}$.

Example Problem

Example : Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5 and 9. What is the form of the general solution?

Solution : Here, there are only 3 roots,

$r_1 = 2$, with multiplicity 3, $r_2 = 5$, with multiplicity 2 and

$r_3 = 9$, with multiplicity 1.

Therefore, the solution is of the form:

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)(2)^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n)(5)^n \\ & + (\alpha_{3,0})(9)^n \end{aligned}$$

Example Problem

Example : What is the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3} \text{ with } a_0 = 1, a_1 = -2, a_2 = -1?$$

Solution : The characteristic equation is $r^3 + 3r^2 + 3r + 1 = 0$.

There is only a single root, $r = -1$, with multiplicity 3.

Therefore, the solution is of the form

$$a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n$$

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n + \alpha_{1,2}n^2)(-1)^n$$

where $\alpha_{1,0}$, $\alpha_{1,1}$ and $\alpha_{1,2}$ are constants.

From the given initial conditions,

$$a_0 = 1 = \alpha_{1,0}, a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2} \text{ and}$$

$$a_2 = -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}.$$

Solving, we find that $\alpha_{1,0} = 1$ and $\alpha_{1,1} = 3$ and $\alpha_{1,2} = -2$.

Hence, $a_n = (1 + 3n - 2n^2)(-1)^n$.